

Literature-implied partial answer: subsymmetric bases with LRP

Run `fa_banach.001`; source arXiv:1705.03924; supporting arXiv:2603.21733

Status

literature_implied_answer (partial subcase).

Albiac–Ansorena–Wallis ask in arXiv:1705.03924, Problem 3.3, whether every Banach space with a subsymmetric basis admits a 1-greedy renorming. This note records the later partial answer supplied by Albiac–Ansorena–Berasategui–Berna.

Claim

Let $\mathcal{X}$ be a subsymmetric basis of a Banach space X . If the upper superdemocracy function, equivalently the fundamental function in this subsymmetric setting, has the lower regularity property, then X admits an equivalent norm for which $\mathcal{X}$ is isometrically greedy, isometrically bidemocratic, and isometrically subsymmetric.

In particular, if X has nontrivial cotype and $\mathcal{X}$ is subsymmetric, then X admits such a 1-greedy renorming.

Identification

The source question asks for a 1-greedy renorming of an arbitrary subsymmetric basis. Theorem 5.2 of arXiv:2603.21733 gives exactly this conclusion under the lower regularity property on the fundamental function. Theorem 5.3 of the same paper derives the nontrivial-cotype subcase by combining Theorem 5.2 with the standard fact that quasi-greedy bases in spaces with nontrivial cotype have lower regularity for the relevant democracy function.

This is not an exact full answer to Problem 3.3. The supporting paper states that, in the absence of a general answer for subsymmetric bases, one must still study the case where the fundamental function does not have the LRP.

Evidence

The source problem appears in arXiv:1705.03924, page 11, Problem 3.3. The supporting statements are arXiv:2603.21733, page 23, Theorems 5.2 and 5.3. The remaining open scope is stated on arXiv:2603.21733, page 26.

The packet includes local copies of both PDFs:

`source_paper.pdf`, `supporting_paper_2603.21733.pdf`.

Scope

The broad question remains open for subsymmetric bases whose fundamental function lacks the lower regularity property. The supporting paper singles out the dual Schlumprecht-space canonical basis as a concrete remaining question.

Search Notes

Local run-index searches for arXiv:1705.03924, Garling sequence spaces, subsymmetric bases, 1-greedy renorming, 1-GUR, Property (A), and greedy bases found no prior packet for this exact subcase. Web searches on 2026-06-15 found arXiv:2603.21733 as the decisive recent supporting paper, but no full solution of the broad source problem.

References

- [1] F. Albiac, J. L. Ansorena, and B. Wallis, *1-Greedy renormings of Garling sequence spaces*, arXiv:1705.03924.
- [2] F. Albiac, J. L. Ansorena, M. Berasategui, and P. M. Berna, *Isometric renormings for greedy bases in Banach spaces, with applications to the Haar System in $L_p[0,1]$, $1 < p < \infty$* , arXiv:2603.21733.